WORKED SOLUTION

STEP I 2010, Question 13

Probability and Statistics • Poisson Waiting Times

STEP I 2010, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q13

The key idea

"The first text arrives between 1 and 2 hours" is the same as "no text in the first hour, but not no text in the first two hours". Each of those is a Poisson probability of zero events, which is a single exponential. Substituting $x = \mathrm{e}^{\lambda}$ then turns everything into a quadratic, and Vieta's formulas (sum and product of roots) do the rest of the work, including for Mildred.

Solution

We take the usual Poisson model: the number of texts George receives in any interval of length t hours is Poisson with mean λt, and counts in non-overlapping intervals are independent. Time is measured from the start of the morning. Since time is continuous, the probability of the first text arriving at exactly 1 or exactly 2 hours is zero, so it doesn't matter whether the endpoints are included.

George's equation

Let T be the time in hours until George's first text. Then $T > t$ exactly when there are no texts in $[0, t]$, so

$$\mathrm{P}(T > t) = \mathrm{P}(\text{Poisson}(\lambda t) = 0) = \mathrm{e}^{-\lambda t}.$$

The event $\{1 < T \leq 2\}$ is $\{T > 1\}$ with $\{T > 2\}$ removed, and $\{T > 2\} \subseteq \{T > 1\}$, so

$$p = \mathrm{P}(1 < T \leq 2) = \mathrm{P}(T > 1) - \mathrm{P}(T > 2) = \mathrm{e}^{-\lambda} - \mathrm{e}^{-2\lambda}.$$

Multiplying through by $\mathrm{e}^{2\lambda}$ gives $p\mathrm{e}^{2\lambda} = \mathrm{e}^{\lambda} - 1$, that is,

$$p\mathrm{e}^{2\lambda} - \mathrm{e}^{\lambda} + 1 = 0,$$

as required.

Two positive values of λ

Put $x = \mathrm{e}^{\lambda}$. Since $\lambda \mapsto \mathrm{e}^{\lambda}$ is a strictly increasing bijection from $(0, \infty)$ to $(1, \infty)$, positive solutions λ correspond exactly to solutions $x > 1$ of

$$px^2 - x + 1 = 0.$$

Here $p > 0$ (it is the probability of an event that happens whenever, say, exactly one text arrives in $[1, 2]$ and none before), so this really is a quadratic.

> **Ask yourself** It is easy to show the roots are real. How do we know both of them exceed 1?

Real and distinct. The discriminant is $1 - 4p > 0$ because $4p < 1$, so there are two distinct real roots $x_1 < x_2$.

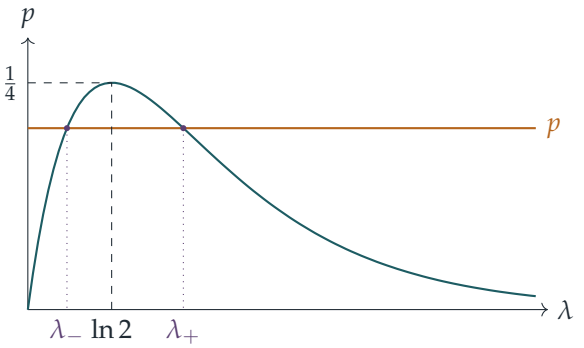
Both exceed 1. By Vieta's formulas, $x_1 + x_2 = \frac{1}{p}$ and $x_1 x_2 = \frac{1}{p}$. Then

$$(x_1 - 1)(x_2 - 1) = x_1 x_2 - (x_1 + x_2) + 1 = \tfrac{1}{p} - \tfrac{1}{p} + 1 = 1 > 0,$$

so $x_1 - 1$ and $x_2 - 1$ have the same sign. Their sum is $x_1 + x_2 - 2 = \frac{1}{p} - 2 > 0$, because $p < \frac{1}{4}$ gives $\frac{1}{p} > 4$. So both are positive: $x_1 > 1$ and $x_2 > 1$.

Hence $\lambda_{\pm} = \ln\left(\dfrac{1 \pm \sqrt{1 - 4p}}{2p}\right)$ are two distinct positive values of λ satisfying the equation.

The picture makes sense of this. As a function of λ, $p = \mathrm{e}^{-\lambda} - \mathrm{e}^{-2\lambda}$ is 0 at $\lambda = 0$, tends to 0 as $\lambda \to \infty$, and (writing $u = \mathrm{e}^{-\lambda}$, $p = u - u^2$) has maximum $\frac{1}{4}$ at $u = \frac{1}{2}$, i.e. $\lambda = \ln 2$. A horizontal line at height $p < \frac{1}{4}$ cuts it twice: a quiet phone (texts rarely arrive at all) and a busy one (texts usually arrive within the first hour) can give the same p.



Mildred: $\lambda_1 + \lambda_2$

Each phone on its own is exactly George's situation, so e^{λ_1} and e^{λ_2} both satisfy $px^2 - x + 1 = 0$. That quadratic has only two roots, and $\lambda_1 \neq \lambda_2$, so e^{λ_1} and e^{λ_2} are its two

roots x_1 and x_2. (This needs $p < \frac{1}{4}$, which holds because two distinct values exist: at $p = \frac{1}{4}$ there is a repeated root.) By the product of roots,

$$e^{\lambda_1+\lambda_2} = e^{\lambda_1}e^{\lambda_2} = x_1x_2 = \frac{1}{p}, \qquad \text{so} \qquad \lambda_1 + \lambda_2 = \ln\frac{1}{p} = -\ln p.$$

Mildred's first text

Ask yourself What is the distribution of the total number of texts Mildred receives on the two phones together?

The two counts are independent Poisson variables, so their sum is Poisson with the sum of the means: in t hours she receives Poisson$((\lambda_1 + \lambda_2)t)$ texts in total. Equivalently, let T_1, T_2 be the waiting times on the two phones; they are independent, and her first text comes at $T = \min(T_1, T_2)$ with

$$P(T > t) = P(T_1 > t)\,P(T_2 > t) = e^{-(\lambda_1+\lambda_2)t}.$$

Either way, writing $\mu = \lambda_1 + \lambda_2$, so that $e^{-\mu} = p$, the same argument as for George gives

$$P(1 < T \leq 2) = e^{-\mu} - e^{-2\mu} = p - p^2 = p(1-p).$$

Checks

- With $\lambda = 0.8$, $p = e^{-0.8} - e^{-1.6} \approx 0.2474$; a simulation of 2×10^5 exponential waiting times gave 0.2462.
- With $p = 0.2$: the roots are $\lambda_1 \approx 0.3235$ and $\lambda_2 \approx 1.2859$, both positive, with sum $1.6094 = \ln 5 = -\ln 0.2$. Simulating $\min(T_1, T_2)$ gave 0.1601 for the final probability, against $p - p^2 = 0.16$.
- Sense check: $p(1-p) < p$, as it should be, since with two phones the first text tends to come sooner, making "no text for an hour" less likely.

Common slips

- Showing the roots in x are real and positive but forgetting that $\lambda > 0$ needs $x > 1$, not just $x > 0$.
- Adding the probabilities for the two phones, or multiplying them (p^2), for the final part. The right event is "first text on either phone", governed by the combined rate.

Beyond STEP

The waiting time to the first event of a Poisson process is exponential, and the merging fact used here (independent Poisson processes with rates λ_1, λ_2 combine into one

with rate $\lambda_1 + \lambda_2$) is one of the basic properties of Poisson processes, alongside "thinning" (keeping each event independently with probability q). These ideas underpin queueing theory and much of applied probability. Grimmett and Welsh, *Probability: An Introduction*, treats Poisson processes in a way that follows on naturally from this question.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.